00 · Reading Map

This folder tells the story of the paper in three acts. Each act is broken into small steps so you can stop after any file and still know where you are.

Act 1: Context (what you need to know first)

File	Question it answers
`01_context_what_agents_retrieve.pdf`	Why does an AI agent need to look things up at all?
`02_context_embeddings.pdf`	What is an embedding, and how does "similar meaning = nearby point" work?
`03_context_the_standard_pipeline.pdf`	What exactly happens, step by step, in the usual "generate then encode" search?
`04_context_hidden_states.pdf`	What are an LLM's hidden states, and why do they matter here?

Act 2: The problems (why the usual way is awkward)

Every problem file follows the same five steps:

1. **What happens** — the behavior in plain words
2. **Why it hurts** — the cost
3. **A concrete example**
4. **How people patched it**
5. **What a real fix would need**

File	Problem
05_problem_overview.pdf	The four problems at a glance and how they connect
06_problem_1_two_models.pdf	Running and maintaining two models
07_problem_2_latency.pdf	The extra step adds delay
08_problem_3_information_bottleneck.pdf	The embedder only sees a short sentence, not the conversation
09_problem_4_why_patches_fall_short.pdf	Why query rewriting, HyDE, hybrid search, and LLM-as-embedder don't remove the split

Act 3: The solution (built one step at a time)

Every solution step file follows the same five steps:

1. **Goal of this step**
2. **What the paper does**
3. **Why it is done this way**
4. **Which problem it answers**
5. **What could go wrong**

File	Solution step	Math file(s) that go with it
10_solution_overview.pdf	The whole idea in one picture	math/00, math/01
11_solution_step_1_the_core_insight.pdf	Reuse what the LLM already computed	—
12_solution_step_2_extract_hidden_states.pdf	Keep the hidden states instead of throwing them away	math/02
13_solution_step_3_the_projection_head.pdf	Turn many LLM vectors into one search vector	math/02 → math/06

File	Solution step	Math file(s) that go with it
14_solution_step_4_learn_from_a_teacher.pdf	Copy an existing embedder so the old index still works	math/07
15_solution_step_5_three_losses.pdf	Why one loss isn't enough	math/08 → math/11
16_solution_step_6_training_and_deployment.pdf	How it is trained and how it runs afterwards	math/12

Act 4: Did it work?

File	Question
17_results_how_it_is_measured.pdf	What do Recall@10, MRR, nDCG, McNemar and confidence intervals mean?
18_results_what_the_experiments_show.pdf	Main result, ablations, training recipe, failures
19_critical_reading_and_limitations.pdf	What the paper does *not* show, and a few numbers worth double-checking